

OPERATIONS MANUAL · AS OF 28 AUGUST 2026

Remote Human-in-the-Loop Control for Multi-Day Claude Code SDLC

*MacBook Pro M4 + iPhone 15 Pro (AT&T) + Spectrum home network
mattpocock skills · 4-stage MVP · 20–60 minute decision cadence*

Audience: a single operator running a 5-day constant full-stack web-dev chat that already auto-compacts at 50% context, uses --dangerously-skip-permissions, and wastes most wall-clock time waiting for HITL answers because the operator is not next to the laptop.

Goal: kick the process off in Ghostty on the MacBook; get visual + audio alerts on the iPhone the moment the session pauses; answer, redirect, or clarify from either device; keep both surfaces in sync in real time.

Sources reviewed 28 Aug 2026: Anthropic Claude Code Remote Control / mobile / Channels docs; Happy (happy.engineering); Moshi comparisons; PocketShell 6-option matrix (Jul 2026); CC Pocket; Herdr; Claude Code GitHub issues #53709, #61177, #55619; mattpocock/skills README and wayfinder docs.

1. Overview and problem restatement

This is not a “keep SSH alive” problem. The session already runs for days with permissions skipped. The stall is almost always a **conversational checkpoint**: a grill question, a wayfinder decision ticket, a prototype pick, a stage-gate authorization, or a redirect after compaction. Average gap is 20–60 minutes. Every unanswered pause burns Max-plan tokens already spent and idles parallel subagents.

The correct product is a **shared control surface**:

- **One live conversation** bound to the local repo on the MacBook (filesystem, MCP, tests, external 4 TB volume stay local).
- **Two equal clients** — Ghostty on the MacBook and an iPhone app — that read and write the same session in near real time.
- **A doorbell that makes noise** on AT&T cellular when the session blocks. Silent banners are a failed design for this cadence.
- **A break-glass shell** when the official bridge hangs, compaction eats the thread, or a wizard needs the Mac GUI.

Constraint that changes the ranking: official Claude iOS Remote Control push has a documented class of silent notifications (no sound / no vibration) even when iOS banners are enabled (claude-code#53709, iPhone 15 Pro). Treat official push as *visual-only unless you verify sound on your unit*. Pair it with a second channel that you know can ring.

A second constraint: a 24 GB M4 MacBook is a laptop, not a Mac Mini. Lid-close sleep, 24 GB pressure from Claude + 4 subagents + browser + Ghostty, and an external OWC volume that can unmount are operational risks equal to the iPhone app choice.

2. Your environment (facts this manual assumes)

Layer	Your kit	Implication
Phone	iPhone 15 Pro, AT&T cellular, good coverage	Use outbound HTTPS / WireGuard. Dynamic Island + Live Activities are available. Focus modes must exclude the doorbell apps.
Home WAN	Spectrum residential	Assume CGNAT or flaky inbound. Do not port-forward SSH. Tailscale or outbound relays only.
Host	MacBook Pro M4, 24 GB unified, 1 TB internal + 4 TB OWC 1M2, macOS Sequoia	Must stay awake on power. Prevent lid-sleep. Keep OWC mounted (do not let the volume sleep-spin down mid-session). 24 GB is tight for many parallel subagents.
Shell	Ghostty + oh-my-zsh + powerlevel10k, Python 3.12 + uv	Ghostty already forwards Claude desktop notifications to Notification Center. Herdr can sit inside Ghostty as the agent multiplexer.
Agent	Claude Code CLI, mattpocock-skills, 5-day chat, compact @ 50%, --dangerously-skip-permissions, 4 MVP stages, parallel subagents	HITL is questions/decisions, not Allow/Deny on every Bash. Permission-button-only apps are insufficient.
Models	Several Anthropic Max \$200/mo; OpenAI \$100; Grok / Gemini / GLM \$20; OpenRouter, Fireworks, Together by API	Official Remote Control requires a claude.ai Max/Pro login — API keys and third-party routers do not drive RC. Other CLIs (Codex, Gemini) need Happy / Moshi / PocketShell.

3. Target architecture and flow

Two planes. The decision plane is how you talk to the 5-day chat. The execution plane is how you keep the laptop reachable and recover when the decision plane lies.

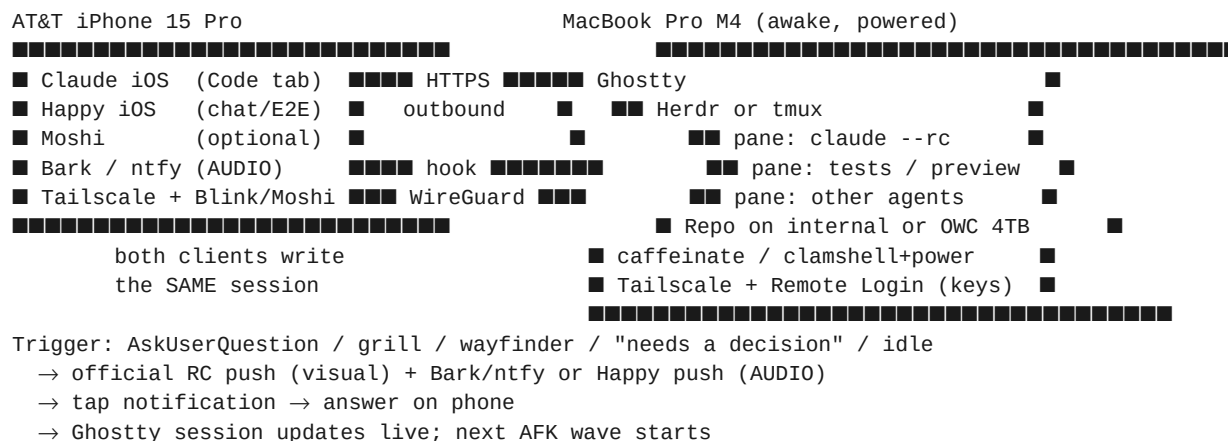


Figure 1. Decision plane (Claude app / Happy) plus execution plane (Tailscale + Herdr/tmux). Audio doorbell is a separate hook.

Checkpoint loop (every 20–60 min)

1. Agent reaches a mattpocock gate (grill branch, wayfinder ticket, prototype pick, stage authorization, post-compact “where were we”). It must call AskUserQuestion or an explicit notify — not bury the ask in prose.
2. Host fires two doorbells: official Remote Control push, plus a Notification hook to Bark or ntfy (sound + critical alert). Happy’s own push can replace the hook if Happy is the primary phone UI.
3. You answer on iPhone or Ghostty. First response wins. Both UIs show the new turn.
4. Agent resumes the current wave / spawns subagents for parallel tickets. You lock the phone.
5. End of day: /handoff into a file + issue tracker. Do not treat a 5-day compacted transcript as the system of record.

4. Corrections to the current 5-day practice

Keep the long-running local agent, compaction at 50%, and Max-plan horsepower. **Change** four habits that fight remote HITL.

Stop using one immortal chat as the only memory.

mattpocock skills assume artifacts: CONTEXT.md, ADRs, wayfinder map issues, vertical-slice tickets, handoff.md. After several 50% compacts, the live transcript is a lossy cache. Phone answers should land in the tracker (“decision: use Stripe Checkout, not a custom billing service”) so tomorrow’s session can reload without you re-explaining.

Split HITL vs AFK inside the same epic.

You already skip tool permissions. That is fine for AFK implement waves. It is the wrong default for grill / wayfinder / architecture. Those pauses are the product. Label tickets **ready-for-afk** vs HITL. Tell the session: on AFK tickets, implement + TDD + stop at code-review; on HITL tickets, AskUserQuestion and wait. Parallel subagents only on unblocked AFK tickets.

Do not skip the notification layer because permissions are skipped.

With `--dangerously-skip-permissions`, official “actions required” pushes fire less often. Your stalls are questions and authorizations. Turn on **Push when Claude decides**, add a standing rule (“notify me and use AskUserQuestion whenever you need a human decision”), and hook `idle_prompt|elicitation_dialog` so silence still rings.

Treat the MacBook as a server for the week.

Power adapter in. Disable lid sleep while on power (Amphetamine, `caffeinate -dims`, or Sequoia Battery settings). Attach OWC and prevent spin-down. Close unused Electron apps. Cap concurrent subagents if memory pressure causes JetEngine / macOS jetsam. Ethernet or 5 GHz Wi-Fi beats flaky Spectrum Wi-Fi for a 5-day RC heartbeat (RC can exit after ~10 minutes awake with no network).

5. Market landscape as of 28 Aug 2026

Three architectures exist. Only relays attached to *your* MacBook satisfy “local repo + 4 TB volume + existing 5-day chat.” Cloud agents (Claude Code on the web, Codex cloud, Cursor cloud) are a different product: they do not inherit your OWC checkout, `uv venv`, or in-flight compacted session.

5.1 First-party Anthropic

Remote Control (Feb 2026, still the default recommendation in Aug docs). Local `claude` process; phone/web are peers. Outbound HTTPS only. Bidirectional sync of conversation, subagents, and workflows. QR / Code tab. Included on Max. API keys, Bedrock, Vertex, custom `ANTHROPIC_BASE_URL` are unsupported — relevant because you also hold OpenRouter / Fireworks keys.

Channels (Telegram, Discord, iMessage plugins). Same local session, hackable async surface, native messenger notifications (these *do* play sound). Weaker rich UI than the Claude app; better doorbell. Good as a pager, not as the only IDE.

5.2 Free / OSS relays (closest to requirement #5)

Happy (happy.engineering, MIT, ~21k stars mid-2026). `npm install -g happy`, wrap the local agent, pair iOS/Android/web by QR. E2E encryption; relay sees ciphertext. Chat UI, steer, approve, review, push, continue the same session in the terminal. Also drives Codex. This is the leading OSS answer to “phone chat that is the same session.”

CC Pocket (App Store, local-first daemon, optional self-hosted relay). Claude Code, Codex, OpenCode, others. Watch, approve, continue, diffs, multi-machine. Good if you refuse a hosted relay.

Orca / MobileFlow / Termopus / Tactic-class apps. Multi-agent or Cloudflare self-host variants. Useful, but smaller or more setup than Happy + official RC.

cc-remote-approval / ntify / Bark hooks. Not a chat UI. They are the reliable audio pager: `PermissionRequest` + `AskUserQuestion` + idle → push with sound. Use beside any chat surface.

5.3 Inexpensive commercial / native iOS terminals

Moshi — free tier; Pro about \$7.99/mo, \$69.99/yr, or \$199 lifetime (Aug 2026 comparison pages). Native Mosh, chat view over your `tmux/Herdr` session, `moshi-hook` inbox, Live Activities / Dynamic Island, Apple Watch approve, diffs, voice. Best iOS terminal for agents if you already have Tailscale.

PocketShell — Free / \$4.99 mo / \$39.99 yr / \$99.99 life. Terminal + resume into live shell + completion push. Not an in-app composer.

Blink Shell — ~\$20/yr. Best raw Mosh client. No agent inbox.

RemoteCode — free on LAN/VPN, ~\$4.99/mo relay. Mac-to-iPhone only.

Omnara — free 10 sessions/mo then ~\$9/mo. Dashboard / Watch. Hosted transcripts in paid mode — worse privacy fit.

5.4 Host multiplexer (MacBook side)

Herdr (herdr.dev, AGPL, ~33k stars, v0.8.x as of mid-Aug 2026). Agent-aware tmux successor inside Ghostty. Sidebar states: blocked / working / done / idle for Claude Code, Codex, Grok CLI, Gemini CLI, OpenCode, and others. Detach/reattach including over SSH. Moshi can pick Herdr sessions. This is the right host companion for 4 parallel waves — tmux cannot tell you which pane is blocked.

cmux is a Ghostty-based native macOS alternative. Stay on Ghostty+Herdr unless you want to leave Ghostty.

6. Decision matrix

Prices and limits checked against public pages dated Jun–Aug 2026. Confirm in-app before paying. “Audio” means the product is known to support a ringing iOS notification when configured; official RC audio is unverified / historically broken.

Option	URL / install	Price	Sync model	Audio HITL alert	Same-session chat	Fits this use
Claude Remote Control	code.claude.com/docs/en/remote-control claude remote-control	Incl. with Max	Outbound HTTPS via Anthropic; Mac + phone + web peers	Visual yes; sound historically unreliable (#53709)	Yes — first-party	Primary decision UI
Claude Channels	Telegram / Discord / iMessage plugins	Incl. with Max	Local agent ↔ messenger	Yes (messenger app sound)	Partial (async messages)	Pager / backup
Happy	happy.engineering npm i -g happy	Free (MIT)	E2E relay; self-hostable	Yes (app push)	Yes — designed for this	Best OSS parallel UI
Moshi + hook	getmoshi.app	Free; Pro ~\$70/yr	Your Tailscale + Mosh + Herdr/tmux	Yes + Live Activity + Watch	Chat view over tty	Best if you want a real shell
CC Pocket	App Store / OSS daemon	Free	Local daemon; optional self-host relay	Yes (app)	Yes (agent-dependent)	Local-first alternative
PocketShell	pocketshell.app	Free–\$99 life	P2P/TURN	Completion push	Resume-to-terminal, not composer	Secondary
Blink + tmux	App Store	~\$20/yr + free tmux	Tailscale SSH/Mosh	No agent push	Raw tty only	Break-glass only
Bark / ntfy hook	hooks.Notification in settings.json	Free	One-way pager	Yes (Critical Alerts possible)	No	Required doorbell
Herdr	herdr.dev / brew install herdr	Free AGPL	Local multiplexer	Host sidebar “blocked”	N/A (host)	Required on 4-wave Mac
Cloud agents	claude.ai/code web, Codex cloud	Plan fees	Vendor VM	Vendor push	Different session	Does not attach to 5-day local chat

7. Top three options

Option A — Official Remote Control + audio hook + Herdr (recommended default)

What it is. Keep the 5-day Claude Code process in Ghostty/Herdr. Attach Remote Control so the Claude iOS app and Ghostty are peers on one session. Add a Notification hook to Bark or ntfy so AT&T delivers a sound. Tailscale + Blink or Moshi exist only as break-glass.

Cost. \$0 extra on top of Max. Bark/ntfy free. Herdr free. Tailscale free personal. Optional Blink ~\$20/yr.

Pros. True bidirectional sync of the actual Claude Code conversation, subagents, and local filesystem — the thing you already have running. No second wrapper process. Outbound-only, Spectrum-safe. Code tab lists “Requires Action.” Max plan already paid.

Cons. iOS push may be silent. Some permission/dialog edge cases still hang (#61177, #55619) — less relevant while skip-permissions is on, still relevant for AskUserQuestion. Local-only slash commands (/plugin, /resume). RC does not drive Codex/Gemini/Grok CLIs. ~10 min host-offline timeout. Session dies if the MacBook sleeps or the OWC unmounts.

Setup (host).

```
# 1. Stay awake
caffeinate -dims &

# 2. Multiplexer
brew install herdr
# inside Ghostty: unbind Ghostty tab/split chords so Herdr owns Cmd+T / Cmd+D
herdr
# one workspace per MVP stage; one pane per wave / subagent family

# 3. Attach RC to the EXISTING 5-day session
# inside the running claude chat:
/remote-control
/config      # remoteControlAtStartup true
              # Push when actions required  true
              # Push when Claude decides    true

# 4. Audio doorbell (Bark example)
# ~/.claude/settings.json hooks.Notification matcher:
#   idle_prompt|elicitation_dialog|permission_prompt|agent_notification
# command: curl Bark / ntfy with sound=alarm

# 5. Standing instruction in CLAUDE.md
# Prefer AskUserQuestion with 2-4 options.
# Notify on every human decision. One decision per turn.
# After compact, restate the open HITL question first.
```

Setup (iPhone). Claude app, same Max account, OS notifications allowed, Focus exception, Critical Alerts on Bark/ntfy. Optional: Tailscale + Blink for herdr attach if RC drops.

Option B — Happy as the parallel phone chat (best OSS if RC UX is not enough)

What it is. Wrap the local Claude (and Codex, if you use the \$100 OpenAI sub in parallel) with Happy’s CLI. iPhone app is a real chat client on the same session, E2E encrypted, with push when the agent needs you. Ghostty remains the host tty.

Cost. Free. Self-host the relay if you do not want their backend to carry ciphertext.

Pros. Built for the exact loop: run, stall, notify, answer from phone, resume same repo/tools/context. Works for Claude + Codex under one app. Does not depend on Anthropic RC feature flags. Strong reviews. MIT.

Cons. Extra daemon/wrapper around a chat you have already kept alive 5 days — attach carefully; test on a throwaway session first. Not a general Unix shell. Relay architecture (even E2E) is a new trust surface versus official outbound RC. You still need the MacBook awake.

When to pick B over A. Official push stays silent on your 15 Pro; you want one phone app for Claude Max + Codex; or RC keeps dropping the 5-day session after compacts.

Option C — Moshi + Tailscale + Herdr + moshi-hook (best if you also need a real shell)

What it is. Herdr holds the 5-day Claude pane plus parallel agent panes. Tailscale reaches the M4 from AT&T. Moshi attaches with Mosh, renders a chat view over that pane, and moshi-hook turns stalls into Dynamic Island + Watch + sounding push.

Cost. Tailscale free personal. Herdr free. Moshi free to start; Pro ~\$70/yr or \$199 life if you want the full agent cockpit.

Pros. Survives Wi-Fi↔LTE. You can type into Claude, run `uv`, inspect the OWC tree, and drive Gemini/Grok/Codex CLIs from one phone. Best audio/visual presence (Island + Watch). Matches a multi-model operator.

Cons. More moving parts. Chat view is a veneer on a tty, not Anthropic’s native Code UI. You must keep Tailscale + Herdr + Mosh healthy. Setup time hours, not minutes.

8. Ranked recommendation and reasoning

Rank 1 — Ship today: Option A. You already pay for several Max seats, you already have a 5-day local Claude Code chat, and requirement #5 is “same session, both devices, alert on pause.” That is the Remote Control spec. Adding Bark/ntfy fixes the known silent-push hole. Adding Herdr makes 4 MVP waves + subagents visible so you do not answer the wrong pane. Adding Tailscale is cheap insurance on Spectrum + a sleeping-lid laptop.

Rank 2 — If A fails the doorbell or the UI in the first 48 hours: add Option B (Happy) rather than replacing A. Happy is the leading free OSS product whose marketing sentence matches your sentence. Run it on a second Herdr pane or wrap a clone of the session after `/handoff`. Do not blindly wrap the 5-day process mid-compact without a snapshot.

Rank 3 — Option C when you are also driving Codex / Gemini / Grok CLIs from the same phone, or when you are tired of fighting RC reconnects on AT&T. Moshi is the leading inexpensive commercial iOS agent terminal as of Aug 2026. It is the right long-term cockpit for a multi-vendor operator; it is not the fastest path to unblocking tonight’s grill question.

Do not rank Telegram-only, public SSH, or cloud VMs first. Telegram/Discord are excellent *paggers* (Channels or cc-remote-approval) and poor places to do mattpocock design decisions. Public SSH on Spectrum is how you get scanned. Cloud agents abandon the OWC checkout and the compacted chat.

Priority	Action	Done when
P0 tonight	In the live chat: <code>/remote-control</code> , enable both push toggles, same-account Claude iOS app, Focus exception. Verify a test <code>AskUserQuestion</code> appears on the phone over AT&T (leave Wi-Fi).	You can send a reply on the phone and see it in Ghostty within seconds.
P0 tonight	Install Bark or ntfy; hook <code>idle_prompt</code> + <code>elicitation_dialog</code> ; enable sound / Critical Alert. Test with phone locked in a pocket.	You hear the stall without looking at the screen.
P0 tonight	caffeinate -dims; power adapter; OWC set not to sleep; confirm lid-close policy.	Closing Messages or walking away does not suspend claude.
P1 tomorrow	brew install herdr; move the 5-day chat into a named Herdr workspace; one pane per MVP wave. Optionally Tailscale + Blink.	Sidebar shows which wave is blocked.
P1 tomorrow	CLAUDE.md: <code>AskUserQuestion</code> + notify rules; compact recovery rule; HITL vs AFK ticket labels; <code>/handoff</code> at end of day.	A compact no longer buries the open question.
P2 this week	Trial Happy on a throwaway repo. If official push stays silent or RC drops after compact, promote Happy to the phone client.	You have a measured fallback, not a theory.

Priority	Action	Done when
P3 optional	Moshi Pro trial + moshi-hook if you need Island/Watch and multi-CLI from the phone.	One iOS app covers Claude + Codex + Grok CLI.

9. Hard constraints and failure modes

- **24 GB.** Four parallel subagents plus a compacted 5-day context plus Chrome will swap. Prefer sequential waves inside a stage, or one implement agent + one review agent. Watch Activity Monitor; if jetsam starts, you will lose the session harder than any iPhone app can save.
- **Laptop sleep.** RC reconnect-on-wake is seconds in current builds; process-killed-by-sleep is fatal. Clamshell only with power + external display or an explicit no-sleep tool.
- **OWC 4 TB.** If the project lives on the external SSD, a USB idle-unmount looks like a vanished repo. Keep the volume awake; prefer the project on internal 1 TB if the OWC link is flaky.
- **Skip-permissions + remote.** You will not get a tap-to-allow safety net. git-guardrails-claude-code hooks are still worth installing so a compact-confused agent cannot force-push from a phone reply.
- **Multi-model keys.** Official RC will not attach to an OpenRouter-routed Claude process. Keep the 5-day SDLC on a Max login. Use Happy/Moshi/Herdr for Codex/Gemini/Grok panes.
- **iOS backgrounding.** The phone does not keep a tty alive. Persistence is on the Mac (Herdr/tmux + claude process). The phone is a client that reconnects.
- **Verify audio on device.** Do not trust vendor copy. Send one test notify over AT&T with the phone locked. If Claude app is silent, Bark/Happy must be in the path before you leave the house.

10. Appendix — standing CLAUDE.md snippet

```
# Remote HITL (operator is often on iPhone)
- When you need a human decision, authorization, or redirect: call AskUserQuestion
  with 2-4 concrete options plus a 5-line blast radius. Then notify.
- Do not hide an open question inside a long status paragraph.
- After /compact or a 50% auto-compact: reprint the single open HITL question first.
- AFK tickets: implement + TDD; stop at code-review; do not start the next ticket.
- HITL tickets (grill, wayfinder grilling/prototype, stage gates): wait.
- Never git push --force, reset --hard, or clean -fd without an explicit typed OK.
- End of day: write handoff.md and update the wayfinder map / issue tracker.
```

Document generated 28 August 2026 for the operator's hardware and the public tooling state on that date. Re-check Happy / Moshi / RC changelogs before a paid commitment. This is an operations manual, not a vendor endorsement.